

TALENTFORGE CSE CHALLENGE POC

Execution Roadmap: Launch by August 14, 2024

Focused MVP: Single Project, Maximum Impact

EXECUTIVE SUMMARY

Launch Deadline: August 14, 2024 (8 Weeks)

Vision: Launch a **single, powerful CSE challenge project** that demonstrates:

-  Students learn through real problem-solving
-  AI auto-validates their work instantly
-  Industry experts verify and score submissions
-  Blockchain badges prove competency
-  HR discovers top talent through verified metrics

Not About Money — About **Proof of Skill**

PART 1: STAKEHOLDER VALUE MATRIX

Value for Each Stakeholder

 STUDENTS

What They Get:

- ✓ Learn by doing (real algorithm problem)
- ✓ Instant AI feedback (3-5 seconds)
- ✓ Industry expert validation (human review)
- ✓ Blockchain badge (permanent proof)
- ✓ Be discovered by HR (no resume needed)

Why This Matters:

"My code is my resume.
No lies. No padding. Just proof."

Metric They Care About:

- Badge earned (binary: yes/no)
- Quality score (0-100)
- Industry expert feedback
- HR views & offers

HOD (Head of Department / College Admin)

What They Get:

- ✓ Real-time student skill dashboard
- ✓ See which students are "industry-ready"
- ✓ Placement data (who got badges, who got hired)
- ✓ Department reputation boost (students coding on platform)
- ✓ Corporate connections (employers viewing students)

Why This Matters:

"Show AICTE & management our students are actually industry-skilled."

Metric They Care About:

- Student participation rate
- Badge earning rate (% who pass)
- Employer interest (views, shortlists)
- Placement outcomes

EMPLOYERS (HR)

What They Get:

- ✓ Pre-verified talent with skills proven
- ✓ Filter by: Badge? Quality score? Expert rating?
- ✓ See actual code submissions (GitHub-style)
- ✓ Direct shortlist (no resume screening needed)
- ✓ Instant matching (our algorithm)

Why This Matters:

"Stop wasting time on screening.
Get verified engineers in 2 weeks."

Metric They Care About:

- Quality of candidates discovered
- Time to hire (vs 2-3 months traditional)
- Conversion rate (applications → offers)
- Salary alignment

COMPANY (Your Platform)

What They Get:

- ✓ Proof that concept works
- ✓ First set of verified talent
- ✓ Case studies for Series A
- ✓ Data on what drives hiring success
- ✓ Network effect (more students → more employers → more students)

Why This Matters:

"One problem, scaled perfectly, beats
100 problems done poorly."

Metric They Care About:

- Students enrolled
- Badge earn rate
- Employer registrations
- Job placements
- Revenue model validation

PART 2: THE ONE PROJECT (Hyper-Focused)

Problem: "Design a Load Balancer"

Why This Project?

- ✓ Real-world relevance (every big tech uses this)
- ✓ Tests multiple skills:
 - System design thinking
 - Trade-off analysis
 - Algorithm implementation
 - Code quality
- ✓ Scalable learning (beginner → advanced versions)
- ✓ Employers love it (all interview load balancing questions)
- ✓ Portfolio credible (not just "solved LeetCode")

Project Structure (One Deep Project)

PROJECT: "Build Your Own Load Balancer"

DIFFICULTY: Intermediate

DURATION: 4-6 hours (can be split)

XP/BADGE VALUE: 1000 XP (highest single badge)

WHAT STUDENTS DO:

- 1. Design load balancing algorithms (30 min theory)
- 2. Implement in Python/Java/C++ (2 hours coding)
- 3. Test against real traffic patterns (1 hour testing)
- 4. Document design decisions (1 hour writing)
- 5. Submit for evaluation

WHAT AI DOES:

- ✓ Code auto-grading (correctness, complexity)
- ✓ Test case validation (throughput, latency)
- ✓ Code quality scoring (style, comments, efficiency)
- ✓ Performance analysis (Big O, memory)

WHAT INDUSTRY EXPERTS DO:

- ✓ Review code quality (2-5 min review)
- ✓ Validate design decisions (feedback)
- ✓ Rate approach (1-5 stars)
- ✓ Add notes (for student learning)
- ✓ Approve for badge (yes/no)

RESULT FOR STUDENT:

- ✓ Badge on profile: "Load Balancer Design Expert"
- ✓ Score: 87/100 (verified by AI + human)
- ✓ Expert feedback: "Clean code, good edge case handling"
- ✓ Visible to all HR (searchable, discoverable)

PART 3: 8-WEEK EXECUTION PLAN (BY AUGUST 14)

Timeline Overview

- | | |
|---------|---|
| Week 1: | Project Setup & Architecture (June 3-9) |
| Week 2: | Backend Core + AI Grading (June 10-16) |

Week 3:	Frontend + Code Submission UI (June 17-23)
Week 4:	Industry Expert Review Module (June 24-30)
Week 5:	Blockchain Badges + Student Dashboard (July 1-7)
Week 6:	Beta Testing (75 users) (July 8-14)
Week 7:	Optimization & Bug Fixes (July 15-21)
Week 8:	Launch Preparation & Live (July 22 - Aug 14)

DETAILED SPRINT BREAKDOWN

SPRINT 1: FOUNDATION (Week 1, June 3-9)

Sprint Goal:

- ✅ Project uploaded, grading engine blueprint ready, basic architecture live

Tasks (Daily Standups Every Day 10 AM):

Monday June 3:

- ☐ Project spec finalized (Load Balancer Design requirements)
- ☐ AWS account setup + CI/CD pipeline
- ☐ GitHub repos created (backend, frontend, contracts)
- ☐ Database schema designed (students, submissions, badges)
- ☐ Team onboarded (5 engineers, 1 PM, 1 designer)

Tuesday June 4:

- ☐ Authentication API (signup, login, OAuth)
- ☐ Student profile schema (college, branch, email verification)
- ☐ Problem details page backend (project description, requirements, examples)
- ☐ Initial Docker setup for code execution

Wednesday June 5:

- ☐ PostgreSQL database deployed
- ☐ API documentation started (Swagger)
- ☐ AI grading module architecture designed
- ☐ Test case framework prepared

Thursday June 6:

- ☐ Code submission endpoint (file upload to S3)

- ☐ Async job queue setup (Bull/RabbitMQ)
- ☐ Initial UI wireframes (5 key screens)
- ☐ Security audit baseline

Friday June 7:

- ☐ Backend testing (all endpoints)
- ☐ Deployment to staging environment
- ☐ Team demo to stakeholders
- ☐ Sprint retrospective

Deliverables:

- ☒ Backend API (60% complete)
- ☒ Database live
- ☒ CI/CD pipeline working
- ☒ 10 endpoints functional
- ☒ Staging environment ready

Success Metrics:

- 0 critical bugs
 - API response time < 200ms
 - Database queries optimized
-

SPRINT 2: AI GRADING ENGINE (Week 2, June 10-16)

Sprint Goal:

- ☒ Code executes, AI grades submissions, scoring system complete

Tasks:

Monday June 10:

- ☐ Docker containers (Python, Java, C++)
- ☐ Code execution sandbox isolated
- ☐ Timeout handling (30-sec max)
- ☐ Memory limits enforced (256MB)

Tuesday June 11:

- ☐ Test case framework (10 hidden test cases)
- ☐ Correctness checker (output validation)
- ☐ Time complexity analyzer (Big O detection)
- ☐ Space complexity calculator

Wednesday June 12:

- ☐ Code quality linter (PEP-8, style checks)
- ☐ Feedback generator (auto-written suggestions)
- ☐ Scoring algorithm (correctness 60%, complexity 30%, style 10%)
- ☐ Results storage (MongoDB)

Thursday June 13:

- ☐ End-to-end testing (5 sample submissions)
- ☐ Performance optimization (grading < 5 seconds)
- ☐ Error handling (execution failures, timeouts)
- ☐ Logging setup (all evaluations logged)

Friday June 14:

- ☐ Staging deployment (complete pipeline)
- ☐ Load testing (100 concurrent submissions)
- ☐ Demo to stakeholders
- ☐ Sprint retrospective

Deliverables:

- ☒ AI grading engine complete
- ☒ 5 test cases validated
- ☒ Scoring algorithm live
- ☒ 99.5% uptime in staging
- ☒ Feedback generation working

Success Metrics:

- Average grading time: 3-5 seconds
 - Accuracy: 100% (test cases pass correctly)
 - No timeouts or memory errors
-

SPRINT 3: FRONTEND + SUBMISSION UI (Week 3, June 17-23)

Sprint Goal:

- ✅ Students can upload code and see instant AI feedback

Tasks:

Monday June 17:

- ☐ Landing page (project description, difficulty, badge preview)
- ☐ Problem details page (requirements, examples, constraints)
- ☐ Code editor UI (Monaco Editor integration)
- ☐ Language selector (Python, Java, C++)

Tuesday June 18:

- ☐ Code submission form (drag-drop or file upload)
- ☐ Real-time validation (syntax check)
- ☐ Submit button with confirmation
- ☐ Loading spinner during grading

Wednesday June 19:

- ☐ Results dashboard (score, breakdown, feedback)
- ☐ Detailed feedback display (colored, highlighted)
- ☐ Test case results (pass/fail, expected vs actual)
- ☐ Performance metrics (execution time, memory used)

Thursday June 20:

- ☐ Resubmit flow (can try again after 7 days)
- ☐ Submission history (all previous attempts)
- ☐ Download badge option (PNG, SVG)
- ☐ Share results (copy link, social media)

Friday June 21:

- ☐ Mobile responsiveness testing
- ☐ Browser compatibility (Chrome, Firefox, Safari)
- ☐ Accessibility testing (WCAG 2.1)
- ☐ Sprint review + demo

Deliverables:

- ☒ Complete frontend UI
- ☒ Responsive design (mobile + desktop)
- ☒ Real-time feedback visible
- ☒ Zero JavaScript errors
- ☒ <2 second page load time

Success Metrics:

- UI tested on 5 browsers
 - Mobile score > 90 (Lighthouse)
 - User testing with 10 students (positive feedback)
-

SPRINT 4: INDUSTRY EXPERT REVIEW (Week 4, June 24-30)

Sprint Goal:

- ☒ Expert evaluation module live, experts can review & score submissions

Tasks:

Monday June 24:

- ☐ Expert portal interface (login, dashboard)
- ☐ Submission queue (show random 5 to review)
- ☐ Code review UI (syntax highlighting, line comments)
- ☐ Rating system (1-5 stars, scoring rubric)

Tuesday June 25:

- ☐ Review form (code quality, design decisions, edge cases)
- ☐ Comments engine (add feedback, suggestions)
- ☐ Approval flow (approve for badge or reject with reason)
- ☐ Time tracking (how long experts spend)

Wednesday June 26:

- ☐ Expert assignment logic (random assignment, 2 experts per submission)
- ☐ Conflict resolution (if 2 experts disagree on score)
- ☐ Expert statistics dashboard (reviews done, average rating, quality)
- ☐ Payment tracking (₹100-200 per expert review)

Thursday June 27:

- ☐ Notification system (alerts when reviews assigned, submitted)
- ☐ Quality checks (flag low-quality expert reviews)
- ☐ SLA tracking (experts review within 48 hours)
- ☐ Integration with scoring system

Friday June 28:

- ☐ End-to-end expert flow testing
- ☐ Security audit (expert data protection)
- ☐ Performance testing (expert dashboard loads fast)
- ☐ Sprint review

Deliverables:

- ☒ Expert portal complete
- ☒ 5 industry experts onboarded & trained
- ☒ Review SLA defined (48-hour turnaround)
- ☒ Quality control in place
- ☒ Expert payments tracked

Success Metrics:

- Expert response rate > 90%
- Review quality score > 4/5
- Time to review < 10 minutes per submission

SPRINT 5: BLOCKCHAIN BADGES + STUDENT DASHBOARD (Week 5, July 1-7)

Sprint Goal:

- ☒ Badges minted on blockchain, student dashboard shows verified credentials

Tasks:

Monday July 1:

- ☐ Smart contract deployed on Polygon Mumbai (testnet)
- ☐ ERC-721 NFT badge standard implemented
- ☐ Batch minting script (for future scalability)
- ☐ Gas cost optimization (<₹10 per badge)

Tuesday July 2:

- ☐ Student wallet integration (Torus, one-click)
- ☐ Badge metadata (project name, score, date, expert names)
- ☐ IPFS upload (code + feedback + certificate)
- ☐ Polygon verification (on-chain proof)

Wednesday July 3:

- ☐ Student dashboard (my badges, scores, expert feedback)
- ☐ Badge display (visual badge, verification link)
- ☐ Certificate generation (PDF, printable)
- ☐ Share to LinkedIn integration (one-click push)

Thursday July 4:

- ☐ Badge gallery (all students' badges visible, anonymized)
- ☐ Blockchain explorer link (view on Polygon scan)
- ☐ Verification API (employers can verify any badge)
- ☐ Revocation mechanism (if plagiarism detected)

Friday July 5:

- ☐ Security audit (blockchain, smart contract)
- ☐ Mainnet readiness (not live yet, testnet only)
- ☐ End-to-end testing (submit → grade → badge mint)
- ☐ Sprint review

Deliverables:

- ☒ Smart contract deployed (testnet)
- ☒ 10+ badges minted successfully
- ☒ Student dashboard complete
- ☒ Blockchain verification working
- ☒ Certificate generation tested

Success Metrics:

- Badge minting time < 2 minutes
- Blockchain transaction success rate 100%
- Certificate downloads working
- Student dashboard load time < 1.5 seconds

SPRINT 6: BETA TESTING (Week 6, July 8-14)

Sprint Goal:

☒ 75 students, 10 HODs, 5 employers testing live platform, feedback collected

Tasks:

Monday July 8:

- ☐ Recruit 75 beta students (email to 10 colleges)
- ☐ Recruit 10 HODs (college admins)
- ☐ Recruit 5 employers (HR) for discovery testing
- ☐ Send invites + onboarding instructions

Tuesday July 9:

- ☐ Beta student cohort 1 submits solutions
- ☐ Gather feedback (Discord channel)
- ☐ Monitor for bugs (real-time)
- ☐ Quick fixes (bug bounty rewards)

Wednesday July 10:

- ☐ HOD testing (can they see student metrics?)
- ☐ Employer testing (can they find students by badges?)
- ☐ Expert review testing (do experts understand UI?)
- ☐ Admin panel testing (can we monitor everything?)

Thursday July 11:

- ☐ Aggregate feedback (what's breaking?)
- ☐ Prioritize fixes (critical vs nice-to-have)
- ☐ Fix top 5 bugs
- ☐ Re-test with beta users

Friday July 12:

- ☐ NPS survey (how likely to recommend? 1-10)
- ☐ Feature feedback (what's missing?)
- ☐ Performance metrics (any slow screens?)
- ☐ Security test (try to break it)

Weekend July 13-14:

- ☐ Post-beta analysis
- ☐ Final bug fixes
- ☐ Prepare launch checklist

Deliverables:

- ☒ 75 students active on platform
- ☒ 50+ submissions received
- ☒ 10+ badges earned (from expert review)
- ☒ 3+ employer accounts active
- ☒ NPS score > 7 (promoter territory)
- ☒ <5 critical bugs remaining

Success Metrics:

- User retention: 80%+ return after first day
 - Submission completion: 65%+ start and finish
 - Expert review time: <12 hours (faster than SLA)
 - Zero payment issues
-

SPRINT 7: OPTIMIZATION & POLISH (Week 7, July 15-21)

Sprint Goal:

- ☒ Platform optimized for scale, all bugs fixed, production-ready

Tasks:

Monday July 15:

- ☐ Performance optimization (page load <1.5 sec)
- ☐ Database query optimization
- ☐ Frontend bundle size reduction
- ☐ CDN setup for static assets

Tuesday July 16:

- ☐ Security hardening (DPDP compliance)
- ☐ Data encryption (PII at rest + in transit)

- ☐ Rate limiting (prevent abuse)
- ☐ SQL injection + XSS prevention verified

Wednesday July 17:

- ☐ Load testing (1000 concurrent users)
- ☐ Stress testing (what breaks?)
- ☐ Auto-scaling validation (can it handle 10x?)
- ☐ Backup & recovery tested

Thursday July 18:

- ☐ UI polish (design consistency)
- ☐ Copy editing (clear messaging)
- ☐ Accessibility audit (screen readers, keyboard nav)
- ☐ Mobile refinement

Friday July 19:

- ☐ Documentation complete (for HODs, students, employers)
- ☐ Onboarding video recorded (3-min walkthrough)
- ☐ FAQ prepared
- ☐ Launch checklist finalized

Weekend July 20-21:

- ☐ Final QA run
- ☐ Backup databases
- ☐ Monitor alerts configured
- ☐ Launch ready

Deliverables:

- ☒ Pageload time < 1.2 seconds (P95)
- ☒ 99.9% uptime test passed
- ☒ All critical bugs fixed
- ☒ SOC 2 controls documented
- ☒ Launch materials ready

Success Metrics:

- Lighthouse score > 90 (all metrics)
- Zero security vulnerabilities

- 0 P1/P2 bugs remaining
 - Load test: 1000 users, 0 errors
-

SPRINT 8: LAUNCH & LIVE (Week 8, July 22 - August 14)

Sprint Goal:

- ☒ Platform live, students enrolling, badges being earned, employers discovering talent

Tasks:

Monday July 22:

- ☐ Production environment go-live
- ☐ Domain goes live (talentforge.io)
- ☐ SSL certificate active
- ☐ Monitoring dashboards live

Tuesday July 23:

- ☐ Email campaign to 500 students (soft launch)
- ☐ Social media posts scheduled
- ☐ LinkedIn announcement from founder
- ☐ Discord community launch

Wednesday July 24 - August 4 (WEEK 1-2 OF LIVE):

- ☐ Daily standup (monitor for issues)
- ☐ Support team responds to emails
- ☐ Students submit solutions
- ☐ Experts review in real-time
- ☐ First badges minted on mainnet
- ☐ Employers start registering

August 5-11 (WEEK 3 OF LIVE):

- ☐ Press release published
- ☐ Case studies written (first 10 student successes)
- ☐ College partnerships announced (MOU signed)
- ☐ Employer testimonials collected

August 12-14 (FINAL PUSH):

- ☐ Public launch announcement
- ☐ All 500 student cohort active
- ☐ 50+ employer signups
- ☐ 100+ submissions reviewed
- ☐ 50+ badges earned
- ☐ Investor demo (show live platform)
- ☐ Success metrics: 4.5+ NPS

Deliverables:

- ☒ Platform live and stable
- ☒ 500+ students enrolled
- ☒ 50+ employers registered
- ☒ 100+ badges earned
- ☒ Press coverage (3+ publications)
- ☒ First placement stories collected

Success Metrics:

- Zero downtime post-launch
- 80%+ student retention (Week 2)
- 65%+ badge earn rate
- 5+ employer shortlists made
- NPS > 40 (students + employers)

PART 4: INFRASTRUCTURE REQUIREMENTS

Must-Have for August 14 Launch

Cloud Infrastructure

AWS Services:

- └ EC2 (t3.medium) – Web server & API
- └ RDS PostgreSQL – Database
- └ S3 – File storage (code submissions)
- └ Lambda – Code execution (serverless)
- └ CloudFront – CDN for static files
- └ Route53 – DNS
- └ CloudWatch – Monitoring & logging

Estimated Cost: ₹25,000/month

Code Execution (Critical Path)

Docker Registry:

- └ Python 3.11 image (100MB)
- └ Java OpenJDK 17 (300MB)
- └ C++ GCC (200MB)
- └ Node.js 18 (150MB)

Kubernetes (auto-scaling):

- └ Min: 2 pods (redundancy)
- └ Max: 20 pods (under load)
- └ Auto-scale on CPU > 70%
- └ Response: 30 seconds

Estimated Cost: ₹15,000/month

Blockchain (Polygon Mainnet)

Smart Contracts:

- └ NFT Badge contract (₹2,000 deploy cost)
- └ Gas fees per badge: ₹5-15
- └ Batch minting (50 at once): ₹200-300 total
- └ Monthly gas budget: ₹10,000

Wallet Service:

- └ Torus (free, open-source)
- └ No hosting cost

Estimated Cost: ₹10,000/month

Database & Storage

PostgreSQL:

- └ 100GB initial
- └ Auto-backup (daily)
- └ 30-day retention
- └ ₹8,000/month

MongoDB (optional, for logs):

- └ Atlas free tier (up to 500MB)
- └ ₹0/month (free tier)

S3 Buckets:

- └ Code storage: 50GB month 1
- └ Cost: ₹200/month

Estimated Cost: ₹8,200/month

Monitoring & Analytics

Datadog:

- └ APM (application monitoring)
- └ Logs (all requests logged)
- └ Alerts (when things break)
- └ ₹10,000/month

Sentry (error tracking):

- └ Real-time error alerts
- └ ₹5,000/month

Estimated Cost: ₹15,000/month

Total Infrastructure Cost

AWS: ₹25,000

Kubernetes: ₹15,000

Blockchain: ₹10,000

Database: ₹8,200

Monitoring: ₹15,000

TOTAL/MONTH: ₹73,200 ($\approx$ ₹75,000/month)

8-Week Cost (1.86 months): ₹1,40,000

PART 5: TEAM STRUCTURE & ROLES

Execution Team (8 People)

1 PRODUCT MANAGER (1)

Role: Project lead, roadmap, stakeholder communication

Hours: Full-time (9 AM - 9 PM IST, daily standups)

Responsibilities:

- Sprint planning (every Monday)
- Stakeholder updates (daily)
- Feature prioritization
- Launch coordination

2 BACKEND ENGINEER (2)

Role: API, database, grading engine

Hours: Full-time

Responsibilities:

- Backend APIs
- Database optimization
- Integration with AI grading
- Expert review system

3 FRONTEND ENGINEER (2)

Role: UI/UX, student dashboard, employer portal

Hours: Full-time

Responsibilities:

- Student submission interface
- Results dashboard
- Expert review UI
- Mobile responsiveness

4 DEVOPS/INFRASTRUCTURE (1)

Role: AWS, Docker, Kubernetes, CI/CD

Hours: Full-time

Responsibilities:

- Infrastructure setup
- Deployment pipeline
- Monitoring & scaling
- Security hardening

5 QA/TESTING (1)

Role: Testing, bug catching, performance validation

Hours: Full-time

Responsibilities:

- Test case creation
- Manual testing
- Load testing
- Beta user coordination

6 BLOCKCHAIN DEVELOPER (0.5 FTE)

Role: Smart contracts, NFT minting

Hours: Part-time (10 hours/week)

Responsibilities:

- Smart contract deployment
- Badge minting automation
- Testnet validation

7 DESIGNER/UX (0.5 FTE)

Role: UI design, brand consistency

Hours: Part-time (10 hours/week)

Responsibilities:

- UI mockups
- Design system
- Brand assets

8 COMMUNITY MANAGER (1)

Role: Discord, support, beta coordination

Hours: Full-time (9 AM - 7 PM IST)

Responsibilities:

- Discord community
- Support tickets
- Beta user communication
- Feedback collection

Total Headcount: 8 FTE Total Monthly Cost: ₹25-30 lakhs 8-Week Total: ₹50-60 lakhs
(including infrastructure)

PART 6: MONITORING DASHBOARDS

For Project Manager (Real-time)

SPRINT HEALTH DASHBOARD:

- └ Sprint velocity (tasks completed)
- └ Burndown chart (work remaining)

- └ Bugs opened vs closed
- └ Code quality metrics (SonarQube)
- └ Deploy frequency
 - └ Team availability/blockers

ACCESSIBLE: Every day, 9 AM standup

TOOL: Jira + Datadog + Custom dashboard

For HOD (Weekly Update)

STUDENT METRICS:

- └ Enrollment: X students from your college
- └ Submission rate: Y% have submitted
- └ Badge rate: Z% earned badges
- └ Average score: 78/100
- └ Time on platform: 4.5 hours average
 - └ Export to Excel (for college reports)

EMPLOYER INTEREST:

- └ Company views: 15 employers viewed your college
- └ Shortlists: 5 students shortlisted
- └ Messages: X recruiter messages sent
 - └ Projected placements: 3-5 students

DEPARTMENT INSIGHTS:

- └ Top performers (top 10 by score)
- └ Skill gaps (what are students struggling with?)
- └ Time investment (who's serious?)
 - └ Recommendations (send specific students challenges)

ACCESSIBLE: Weekly email summary + portal

TOOL: Custom HOD dashboard

For Employers (Real-time Discovery)

CANDIDATE DISCOVERY:

- └ Filter by: Badge earned? Score > 80? College?
- └ Sort by: Latest badges, highest scores, most active
- └ View candidate: Score, badge, expert feedback, code
- └ Quick action: Shortlist, message, schedule interview

COHORT ANALYTICS:

- └ How many badges earned this week?
- └ Average quality score
- └ Comparison to previous cohort
- └ Time to completion (faster trends?)

PIPELINE TRACKING:

- └ Shortlisted candidates: 15 (status: pending interview)
- └ Offered: 3 (status: pending decision)
- └ Hired: 1 (confirmed)

ACCESSIBLE: 24/7 on portal

TOOL: Employer dashboard

For Students (Progress Tracking)

MY JOURNEY:

- └ Project status: In progress / Submitted / Under review
- └ AI score: 87/100 (real-time)
- └ Expert review: Pending / Approved / Needs revision
- └ Badge status: Earned on [date] / Minted on Polygon
- └ LinkedIn: Share badge (one-click)

MY FEEDBACK:

- └ AI's auto-score: Correctness 90%, Complexity 80%, Style 75%
- └ Expert's review: "Great code! Consider edge cases for..."
- └ Time spent: 4.5 hours
- └ Compared to cohort: Top 15%

NEXT STEPS:

- └ Resubmit? (wait 7 days)
- └ Another challenge? (coming soon)
- └ Apply for job? (based on badge, get matched)

ACCESSIBLE: 24/7 on web/app

PART 7: GO-LIVE CHECKLIST (August 14)

2 Weeks Before (Aug 1-7)

TECHNICAL:

- ☐ Performance testing (1000 concurrent users)
- ☐ Security audit (penetration test)
- ☐ Database backup tested (recovery successful)
- ☐ Monitoring alerts configured
- ☐ Runbook written (if things break, here's what to do)

CONTENT:

- ☐ Help documentation (for students, HODs, employers)
- ☐ FAQ page live
- ☐ Onboarding video (2-3 min)
- ☐ Success criteria defined (what counts as "launch success?")

INFRASTRUCTURE:

- ☐ CDN warmed (assets cached)
- ☐ DNS propagated
- ☐ SSL certificate verified
- ☐ API rate limiting active
- ☐ DDoS protection enabled

STAKEHOLDER PREP:

- ☐ Students invited (500 emails sent)
- ☐ HODs invited (50 colleges notified)
- ☐ Employers invited (10 HR contacts)
- ☐ Press release written
- ☐ Founder talking points prepared

1 Week Before (Aug 8-13)

SOFT LAUNCH:

- ☐ 25 early-access students active
- ☐ Verify: signup works, submission works, grading works
- ☐ Verify: expert reviews received, badges minted
- ☐ Fix any issues found

FINAL QA:

- ☐ All critical user flows tested
- ☐ All browsers tested (Chrome, Safari, Firefox)
- ☐ All mobile devices tested (iPhone, Android)
- ☐ Payment flow tested (if applicable)
- ☐ Database backup completed

TEAM READINESS:

- ☐ Support team trained (how to handle issues)

- ☐ On-call rotation established
- ☐ Escalation process defined
- ☐ Incident response plan ready

Launch Day (August 14)

MORNING (6 AM):

- ☐ Final database backup
- ☐ Monitoring dashboards live
- ☐ On-call team standing by
- ☐ Slack channel for real-time updates

AFTERNOON (12 PM):

- ☐ 🇺🇸 Public launch announced
- ☐ Email to 500 students
- ☐ Social media posts published
- ☐ Website live
- ☐ Discord server open

EVENING (6 PM):

- ☐ Monitor for issues
- ☐ Support team responding to tickets
- ☐ Collect feedback (loving it? issues?)
- ☐ First submissions rolling in

POST-LAUNCH (24H):

- ☐ Confirm: 100+ students signed up
- ☐ Confirm: 20+ submissions received
- ☐ Confirm: AI grading working
- ☐ Confirm: Experts reviewing
- ☐ Confirm: Badges starting to mint

PART 8: SUCCESS METRICS (August 14 - August 30)

Launch Week Targets

STUDENT SIDE:

- └ Total signups: 500
- └ Submissions received: 100+ (20% of cohort)
- └ Completion rate: 65% (started and finished)

- └ Average time to complete: 4.5 hours
- └ Badge earn rate: 60% (60 badges earned)
- └ Avg quality score: 75/100

EXPERT SIDE:

- └ Experts active: 5 (all reviewing)
- └ Reviews completed: 60 (for 60 submissions)
- └ Average review time: 8 minutes
- └ Expert quality score: 4.2/5 stars
- └ Zero delayed reviews (SLA: <48 hours)

EMPLOYER SIDE:

- └ Employer registrations: 10
- └ Candidates discovered: 30
- └ Shortlists made: 5
- └ Interview invitations: 2-3
- └ First offers: 0-1 (too early, but monitoring)

PLATFORM HEALTH:

- └ Uptime: 99.9%
- └ Page load time: <1.5 seconds
- └ API latency: <200ms
- └ Grading latency: <5 seconds
- └ Zero critical bugs post-launch
- └ NPS score: >40 (promoter territory)

First Month (August 14 - September 14) Roadmap

WEEK 1-2 (Launch):

- └ Stabilize platform
- └ Collect student feedback
- └ Fix bugs found in wild
- └ Celebrate first badges!

WEEK 3-4 (Growth):

- └ Expand to 1,000 students
- └ Launch referral program
- └ First employer placements
- └ Case studies documented

MONTH 2 (Optimization):

- └ Move to mainnet (Polygon production)
- └ Introduce second challenge

- └ Partnerships with more colleges
- └ Series A preparation deck

PART 9: RISK MITIGATION

Top 5 Risks

RISK 1: Code Execution Failures

- └ Mitigation: Extensive Docker testing pre-launch
- └ Backup: Manual review fallback
- └ Owner: DevOps engineer
- └ SLA: Fix within 2 hours

RISK 2: Expert Review Bottleneck

- └ Mitigation: 5 experts hired + trained pre-launch
- └ Backup: AI score used as temporary approval if expert delayed
- └ Owner: Community manager
- └ SLA: Review within 48 hours

RISK 3: Student Dropout (Low Engagement)

- └ Mitigation: Gamification (badges are motivating)
- └ Backup: Personal emails from HOD encouraging
- └ Owner: Community manager
- └ Target: 80%+ retention Week 1 → Week 2

RISK 4: Blockchain Transaction Failures

- └ Mitigation: Testnet thoroughly tested
- └ Backup: Store badge data in DB, mint on blockchain retroactively
- └ Owner: Blockchain developer
- └ SLA: Resolve within 4 hours

RISK 5: Employer Interest Low

- └ Mitigation: Pre-recruit 5 pilot employers
 - └ Backup: Cold-email 100+ HR contacts simultaneously
 - └ Owner: Product manager
 - └ Target: 10+ employers registered Week 1
-

PART 10: LAUNCH NARRATIVE (For August 14 Announcement)

Press Release

"TalentForge Launches India's First
Performance-Verified CSE Challenge Platform

Students Prove Skills Through Real Problems.
Employers Discover Pre-Verified Talent.
No Resumes. Just Proof.

[CITY], AUGUST 14, 2024 – TalentForge,
an AI-powered talent verification platform,
launches its first challenge:
'Build a Load Balancer' (difficulty: intermediate).

500 students from 50+ colleges are already solving it.
Industry experts are reviewing their code.
Top performers are getting blockchain-verified badges.

Employers can now discover students by skill metrics,
not resume keywords.

First badges earned: 50+
First employer interest: 10+ companies
First interview invitations: 2-3
First placements: Coming Week 2

This is not recruitment.
This is proof."

TARGET PUBLICATIONS:

- └ Economic Times (startup section)
 - └ Moneycontrol (careers/education)
 - └ YourStory (founder story)
 - └ TechCrunch India (product launch)
 - └ LinkedIn (organic reach)
-

PART 11: FINAL TIMELINE SUMMARY

TODAY

AUGUST 14 LAUNCH

|

|

Jun 3

Jun 10

Jun 17

Jun 24

Jul 1

Jul 8

Jul 15

Jul 22

AUG 14

|

|

|

|

|

|

|

|

|

SPRINT 1

SPRINT 2

SPRINT 3

SPRINT 4

SPRINT 5

SPRINT 6

SPRINT 7

SPRINT 8

🎉

ARCH

AI

UI

EXPERT

BADGES

BETA

OPTIMIZE

LAUNCH

LIVE

GRADING

SUBMIT

REVIEW

+ NFT

TEST

POLISH

WEEK

✓ Foundation

✓ AI Grading

✓ Frontend

✓ Expert Module

✓ Blockchain

✓ Tested

✓ Optimized

✓ Ready → LIVE

LIVE METRICS (Target by Aug 14):

└ 500 students enrolled

└ 100+ submissions graded

└ 60 badges earned

└ 10 employers active

└ 5 interviews scheduled

└ 4.5+ NPS

└ 99.9% uptime

PART 12: DELIVERABLES AT LAUNCH (AUGUST 14)

FOR STUDENTS:

✓ Platform (web + mobile responsive)

✓ Signup & authentication

✓ Challenge: "Build a Load Balancer"

✓ Code editor (Monaco)

✓ AI auto-grading (3-5 seconds)

✓ Expert review (human validation)

✓ Blockchain badge (on Polygon)

✓ Share to LinkedIn

✓ Leaderboard (see top performers)

- ✓ Certificate (downloadable)

FOR HODS:

- ✓ College admin portal
- ✓ Student metrics dashboard (real-time)
- ✓ Employer interest tracker
- ✓ Placement tracking
- ✓ Export reports

FOR EMPLOYERS:

- ✓ Discovery portal
- ✓ Filter & search students
- ✓ View code + scores + expert feedback
- ✓ Shortlist candidates
- ✓ Message students
- ✓ Pipeline tracking

FOR PLATFORM:

- ✓ Analytics dashboard
- ✓ Monitoring & alerts
- ✓ Support system
- ✓ Payment processing (future)
- ✓ Admin controls

EXECUTION STARTS NOW.

LAUNCH: AUGUST 14, 2024.

ONE CHALLENGE. INFINITE IMPACT.